Guild Assay Office
Notice of Material Quality
Likely Insider Sabotage/Mutiny/Corruption

October 3, 2026

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1 To All Engineering Cadres of House Veridian

The zhalite smelters of House Veridian have delivered the structural stock for the **Drakkar Ascent Vehicle**. Guild assayers have completed their inspection of every batch of metal plate, composite prepreg, and filament-wound tube destined for the thrust barrel.

The stock is **certified, but only to B-basis (Grade II material). It seems to be a case of insider sabotage, or a case of mutiny. Other Houses of the Imperium might have orchestrated this to bring down the mighty House Veridian!**

Trace impurities have been found in every batch:

- **Metals** — oxide inclusions, slag entrapment, and micro-porosity from the Drakkar smelters. These do not soften the lattice, but they act as stress concentrators and imperfection initiators.
- **Composites** — resin porosity, incomplete fibre wet-out, and fibre volume fraction shortfall from the desert curing ovens. These reduce both stiffness and strength, and they are particularly damaging to thin-shell buckling performance.

The Guild will **not** certify A-basis allowables for this stock. All cadres must re-evaluate their thrust barrel designs using the revised material properties in this notice.

2 What Has Changed Physically

Property	Metals	Composites
Density ρ	Unchanged	Unchanged
Young's modulus E	Essentially unchanged	Reduced $\sim$5%
Strength allowables	Reduced $\sim$20%	Reduced $\sim$20%
Buckling knockdown factor k	$0.50 \rightarrow 0.40$	$0.65 \rightarrow 0.50$

The Critical Insight

- **Global column buckling margins** are almost unaffected, because E and I are unchanged.
- **Local shell buckling margins** collapse, because the knockdown factor is more severe.
- **Material failure margins** collapse, because the allowables are reduced.

Designs that were governed by **global buckling** will barely change. Designs governed by **local shell buckling or material failure** will require thicker walls, more ring frames, or a material change.

3 Revised Material Properties

All cadres must use the following B-basis properties in place of the clean A-basis values issued with the original problem statement.

3.1 Metallic Materials (B-Basis, Impurity-Degraded)

Material	E **(GPa)**	σ_y **(MPa)**	ρ **(kg/m^3)**	ν	**Knockdown** k
Aluminium 7075-T6	71.7	**402**	2810	0.33	**0.40**
Titanium Ti-6Al-4V	113.8	**704**	4430	0.31	**0.40**
Steel 4130	205	**368**	7850	0.29	**0.40**
Magnesium AZ31B	45	**176**	1770	0.35	**0.40**

3.2 Quasi-Isotropic Composite Laminates (B-Basis, Porous)

Composite System	**Eff.** E **(GPa)**	ν	X_c **(MPa)**	Y_t **(MPa)**	S **(MPa)**	ρ **(kg/m^3)**	k
Carbon/Epoxy QI	**52.3**	0.30	**480**	**400**	**160**	1600	**0.50**
Glass/Epoxy QI	**23.8**	0.30	**240**	**200**	**80**	1900	**0.50**
Aramid/Epoxy QI	**33.3**	0.34	**160**	**240**	**64**	1400	**0.50**

Composite allowables remain first-ply failure values under uniaxial loading. Hybrid designs are still permitted, but the analysis must be transparent and must use the B-basis values for each constituent material.

4 Required Deliverable: Guild Assay Response

Each cadre must submit, alongside their original final report, a **Guild Assay Response** of no more than two pages containing:

1. **Revised property set used.** A table showing the B-basis values and knockdown factors applied to your chosen material(s).
2. **Re-check of all margins.** Yield or composite failure, global column buckling, and local shell buckling, using the revised properties. State each margin numerically.
3. **Verdict.** PASS or FAIL, with the governing failure mode and its margin clearly identified.
4. **Minimum mass fix.** If FAIL, propose the lightest modification that restores all margins. Options include increasing wall thickness, adding ring frames, changing material, or any combination. Report the new mass and the mass penalty:

$$\Delta m = m_{\text{fixed}} - m_{\text{original}}$$

5. **Failure mode declaration.** State whether the governing mode *changed* from the clean-material case, and why.
6. **Sensitivity insight.** One paragraph: is your design more sensitive to material quality or to gust strength? Justify with numbers.

5 Revised Scoring

Let M_i be the **Material Integrity Score**:

$$M_i = \begin{cases} 1.0, & \text{passes under B-basis with no modification} \\ 0.7, & \text{fails, fixed with } \Delta m \leq 10\% \\ 0.4, & \text{fails, fixed with } \Delta m \leq 25\% \\ 0.0, & \text{fails, cannot be fixed within } 25\% \end{cases}$$

Let H_i be the **Failure Mode Shift Bonus**:

$$H_i = \begin{cases} 1.0, & \text{correctly states whether the governing mode changed} \\ 0.5, & \text{partially correct} \\ 0.0, & \text{otherwise} \end{cases}$$

The revised total score is:

$$\text{Total}_i = 100 \times [0.35\, S_i + 0.15\, M_i + 0.10\, H_i + 0.25\, R_i + 0.10\, T_i + 0.05\, I_i]$$

where S_i, R_i, T_i, and I_i retain their original meanings.

Justice Shall be Delivered, but Zhalite can't wait!

The Guild Assay Office thanks you for your compliance.